



breakout session
data access and data quality

introduction
Petr Kolínský

AdriaArray Workshop, March 11-14, 2025, San Servolo



ISTITUTO NAZIONALE DI GEOFISICA E VULCANOLOGIA



INSTITUTE OF GEOPHYSICS
OF THE CZECH ACADEMY OF SCIENCES



Orfeus

agenda of today's session

Petr Kolínský:	intro
Frederik Tilmann:	S3 buckets
Luděk Vecsey:	metadata tests
Peter Danecek:	EIDA
Felix Eckel:	workshop on data download

ORFEUS documentation

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COMMUNITY SERVICES:
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GEO-INQUIRE PRODUCTS
Integration of UT network (Ukraine) in EIDA
InfraSound integration guidelines
EIDA Guidelines for the creation of derived products from raw DAS data
ORFEUS & Geo-INQUIRE Workshops 2024

ADRIAARRAY INITIATIVE:
AdriaArray
AdriaArray News

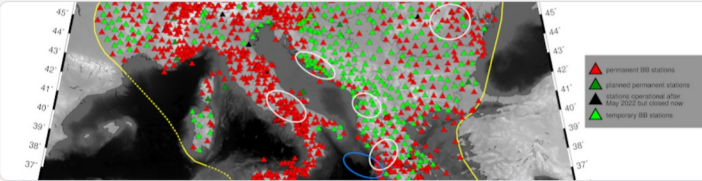
AdriaArray - Organization
AdriaArray - MoC
AdriaArray - Members
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AdriaArray - application
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AdriaArray - Working Groups
AdriaArray - Data analysis and research
AdriaArray - Seismic networks
AdriaArray Communication & Outreach

AdriaArray - Participants

	Country	Name
	Romania	Alina Coman
	Germany	Amr El-Sharkawy
	GREECE	Anastasia Kiratzi
	Italy	Andrea Morelli
	Germany	Andreas Rietbrock
	Greece	Andreas Tzanis
	Romania	Andreea Cralu
	Romania	Andreea Tolea
	Romania	Andrei Balu
	Romania	Andrei Mihai
	Slovakia	Andrej Cipciar
	Romania	Anica Otilia Placinta
	Albania	Anila Xhahysa
	Greece	Anna Serpetidisaki
	Germany	Anne Mohr
	Schweiz	Anne Obermann
	Germany	Antje Schlömer
	Albania	Ardian Minarolli
	Albania	Ardit Dervishi
	Greece	Athanasios Lois
	Germany	Ayoub Kaviani
	Hungary	Barbara Czece
	Germany	Bernd Schurr
	Albania	Besian Rama
	Romania	Bogdan Enescu
	Romania	Bogdan Grecu
	Romania	Bogdan Zaharia
	Hungary	Bálint Süle
	Italy	Carlo Glunchi
	Romania	Catalin Gheabtau
	Italy	Caterina Montuori
	Montenegro	Čavelić Marin
	Greece	Chrisanthi Ventouzi
	Germany	Christian Haberland
	Sweden	Christian Schiffer
	Germany	Christian Weidle
	Greece	Christos Evangelidis
	Italy	Claudia Piromallo
	Austria	Clément Estève
	Romania	Constantin Ionescu
	FRANCE	Coralie Aubert
	Greece	Costas Papazachos
	Romania	Cralu George Marius
	Romania	Cristian Ghita
	Romania	Cristian Neagoie
	Czech Republic	Cédric P. Legendre
	Albania	Damiano Koxhij
	Italy	Damiano Pesaresi
apest, Hungary	Hungary	Daniel Kalmar
	Germany	Daniel Köhn
	Romanian	Daniel Nistor Paulescu
	Slovenia	Danijeta Birkic
	Switzerland	Dario Jozinović
	Greece	Dimitrios Giannopoulos

who wants to be on
the webpage,
fills the
questionnaire



Questionnaire for participants of the AdriaArray Seismology Group (June 2024)

Contact us at: petr.kolinsky@ig.cas.cz

petr.kolinsky@volny.cz [Přepnout účet](#)

* Označuje povinnou otázku

E-mail *

Váš e-mail:

[Další](#) Strana 1 z 8 [Vymazat formulář](#)

Nikdy přes Formuláře Google neposílejte hesla.

Obsah není vytvořen ani schválen Googlem. - [Smluvní podmínky služby](#) - [Zásady ochrany soukromí](#)

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Google Formuláře



ORFEUS is the European Infrastructure for seismic waveform data in **EPOS**.

ORFEUS services are integrated in the **EPOS Data Portal**.

ORFEUS encourages the adoption of open and **FAIR** seismic data policies.

ORFEUS supports the **AdriaArray** initiative.

ORFEUS more information

ORFEUS news and announcements

EIDA Issue Tracker

European Integrated Data Archive EIDA

EIDA, an initiative within ORFEUS, is a distributed federation of datacenters established to securely archive seismic waveform data and metadata gathered by European research infrastructures, and provide transparent access to data for the geosciences research communities. EIDA's **organization and management** is handled by the EIDA Management Board. The **EIDA nodes** are data centres that collect and archive data from seismic networks deploying broad-band sensors, short period sensors, accelerometers, infrasound sensors, and other geophysical instruments.

Seismic networks that participate in EIDA are listed as **contributing networks**.



Webinterface

Graphical Interface for waveform and metadata access.



Webservices

APIs for data and metadata access.



Data Quality

Interfaces for data quality visualization.



Station Book

Access to the entire EIDA station inventory.



Station Book

history of the ORFEUS User Advisory Group

2018, July 30, updated ORFEUS governance:

The products and services delivered by ORFEUS will be planned, assessed and improved through the technical and scientific feedback of a newly established **User Advisory Group (UAG)**, comprised of active European Earth scientists with expertise encompassing a broad range of disciplines.



UAG reports to ORFEUS

1 st report	SM	November 2018 BB January 2019	data access, portals data availability
2 nd report	SM	February 2020 BB January 2020	follow-up on the previous data availability + quality
3 rd report	BB+SM	October 2021	(meta)data quality tools
4 th report	BB+SM	September 2023	metadata quality tools
5 th report	BB+SM	February 2025	various recommendations

Report of the ORFEUS User Advisory Group on actions and tasks taken in 2024

the fifth UAG report to the ORFEUS ExeCom – 7 February, 2025

prepared and written by the UAG members

Dino Bindi, Felix Eckel, Petr Kolínský, Valerio Poggi, Antje Schlömer, Davor Stanko and Luděk Vecsey

Introduction

This report proposes ideas, suggestions and formulates questions by the UAG in 2024. UAG members met in person at several occasions during the year, mainly individually or in a smaller number. A common agenda was focused on publishing the (meta)data test results. Otherwise, the UAG members had their individual agendas, which are compiled in this report. Many points from the previous report were solved or clarified. There are some tasks and suggestions pending. We do not repeat all of them here and we refer to the previous report. The UAG members have been communicating with the personnel of ORFEUS and its bodies throughout the year continuously and many technical questions and issues have been solved on the fly.

Acknowledgement

UAG thanks for the replies to the previous report. UAG acknowledges the successful functioning of the ORFEUS grants (software, data integration, newly mobile pools and community support). UAG shares the thanks from the community for supporting events such as the AdriaArray Sofia 2024 workshop and for organizing the ORFEUS workshop in Helsinki. The online survey was launched, which is appreciated. The ORFEUS Forum turned out to be a useful tool, which resulted in requesting more categories to be added. UAG acknowledges the launch of the availability webservice as well as the preparation of the metadata change logging system.

Collaboration with EarthScope

UAG supports the actions taken for the coordination between ORFEUS and EarthScope. We feel that coordination on a global level is desired.

suggestion #1

The UAG supports the possibility, anticipated by the ORFEUS Secretary General, of organising a joint COSMOS-ORFEUS workshop focused on mapping the landscape of waveform processing strategies in earthquake data centres worldwide, with emphasis on the use of AI modules embedded in the traditional workflows of seismological and engineering seismology data centres. The use of an automatic processing scheme based on standards shared by the scientific community, well documented and tested, could facilitate the inclusion in ESM also of large volumes of velocimetric records, useful in particular for areas with low seismicity (e.g., for the stable continental regions of Europe), analogous to what has been experimented in the Italian database (https://itaca.mi.ingv.it/ItacaNet_40/#/products/itacaext_flatfile)

Results of the (meta)data quality tests

As foreseen in the previous report, and requested by the community, the UAG members made the results of their tests publicly accessible on GitHub: <https://github.com/PetrColinSky/DataQuality/tree/master/summary>. At the moment, the focus is on the AdriaArray stations. The main outcome is a single excel sheet with all stations listed in rows, and results of particular tests given in columns. For easier orientation, the cells are colored. There is a detailed readme file on GitHub, with links to other readme files for particular tests. This summary was introduced by a poster at EGU in April 2024: https://github.com/PetrColinSky/DataQuality/blob/master/summary/EGU24poster_Kolinsky.pdf and <https://meetingorganizer.copernicus.org/EGU24/EGU24-12520.html>. We have a procedure for updating the test summary. Anytime a new test is performed, we create a new summary table. The old one remains in the repository. Hence, any network operator can check the stations of his interest and see both the latest as well as the previous results. The summary of the (meta)data tests was developed by the UAG members together with Johannes Stampa (Uni Kiel) and Tena Belinić Topić (Uni Zagreb). The online results were shared on the ORFEUS Forum <https://forum.orfeus-eu.org/t/adriaarray-news-june-2024/1059> as well as on the EIDA issue tracker <https://github.com/EIDA/userfeedback/issues/166>. There was an immediate response by several EIDA/network operators, thanks to these. However, many issues stayed unsolved by the other network operators.

suggestion #2

UAG asks ORFEUS for support by informing and spreading the news that the tests are available and repeatedly updated and that the station/network/EIDA operators should check it out and act accordingly.

open question

As the response to correct the metadata is slow, it looks like there is a need to continue with testing and updating the result summary. Some of the UAG members involved will be leaving UAG soon. They can still collaborate, however, with limited time only.

Metadata change log

UAG appreciates the work that has been done by Simon Schneider and Jarek Bienkowski regarding the metadata change log, running in a test mode at KNMI as ChangeLog API. We look forward to seeing it released in early 2025.

Availability webservice

The FDSN Availability Service is an excellent tool for streamlining bulk data downloads from EIDA. It allows users to focus on downloading only the data available in EIDA. Additionally, if a download fails, the Availability Service can help verify whether the issue was due to data unavailability or a download/EIDA issue. For the service to be widely adopted, it needs to be both reliable and user-friendly. Regarding reliability, the database of the service is not always up-to-date. For instance, when data is retroactively added to EIDA, it appears that the service's database updates must be triggered manually. This can result in long delays or even overlooked updates. Furthermore, its integration with the EIDA WFCatalog update process is unclear and may introduce additional dependencies.

suggestion #3

Automating the update process and ensuring its independence from WFCatalog.

suggestion #4

Improving usability could involve integrating the availability service into ObsPy subroutines or providing a simple, well-documented Python code example to demonstrate how this service can be utilized in data download scripts. Promoting these tools and examples would further encourage users to adopt the service.

results of the (meta)data quality tests

- the tests are “published” on GitHub, available to everybody
- simplified overview

suggestion #2

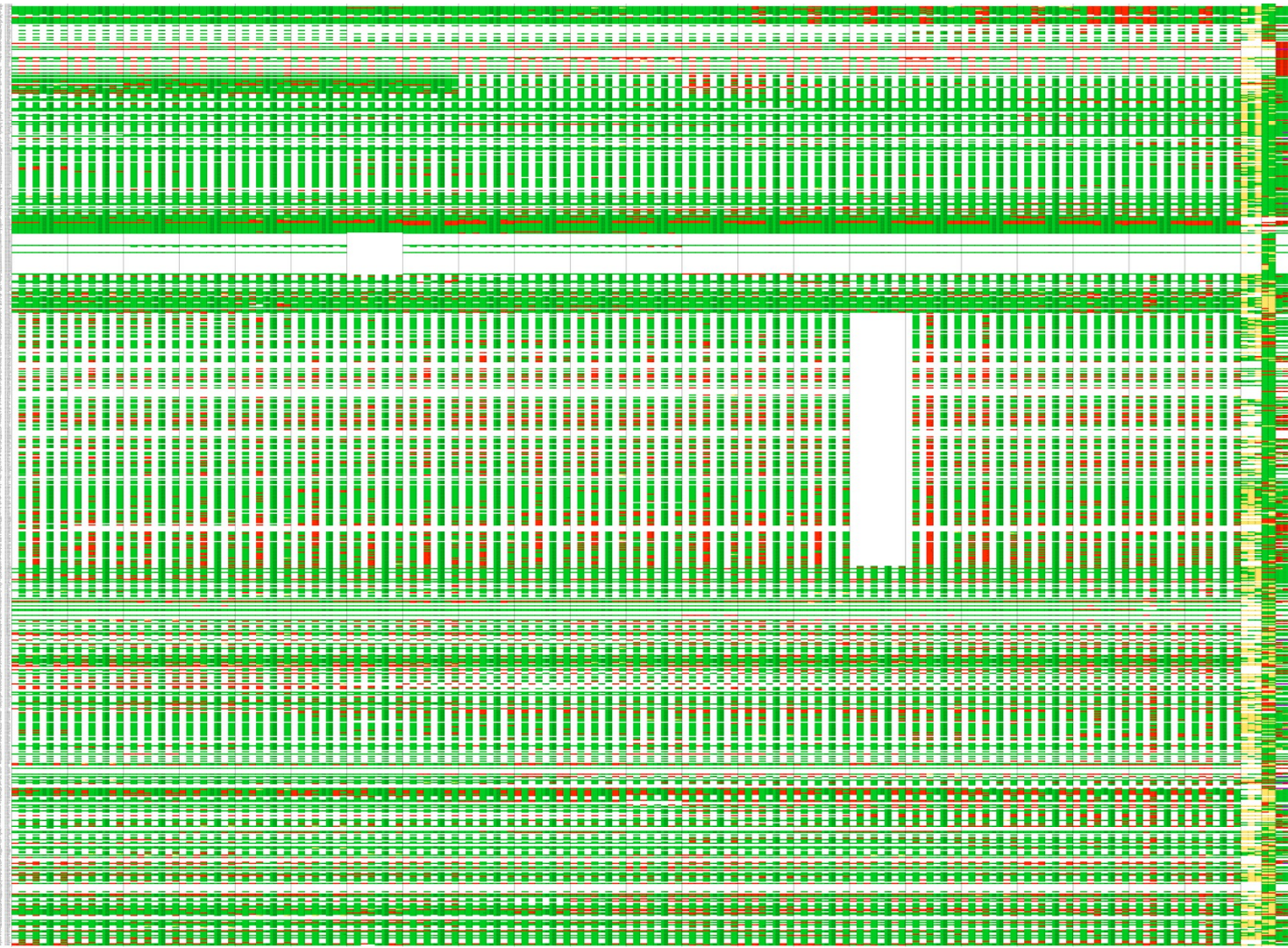
UAG asks ORFEUS for support by informing and spreading the news that the tests are available and repeatedly updated and that the station/network/EIDA operators should check it out and act accordingly.

mass data downloads

suggestion #9

Although the issue might not be connected to the EIDA services but rather to the way users download the data (with ObsPy), a recommendation on how to use the ObsPy client so that the user gets all the data at once would be appreciated.

summary table



disclaimer

- quality of data and metadata is to be ensured by the station/network operators
- individual researchers, as well as the ORFEUS UAG, provide tests of data & metadata, however, in no way it is their responsibility that metadata contains errors, or that data is not available
- people do the tests voluntarily themselves
- members of the UAG also work voluntarily

Methods for data and metadata quality tests of large dense seismic networks - focus on AdriaArray

Petr Kolínský^{1*}, Johannes Stampa^{2,3}, Luděk Vecsey¹, Felix Eckel², Tena Belinić Topić⁴, Thomas Meier² and the AdriaArray Seismology Group⁵

⁽¹⁾ Institute of Geophysics of the Czech Academy of Sciences, Prague, Czech Republic

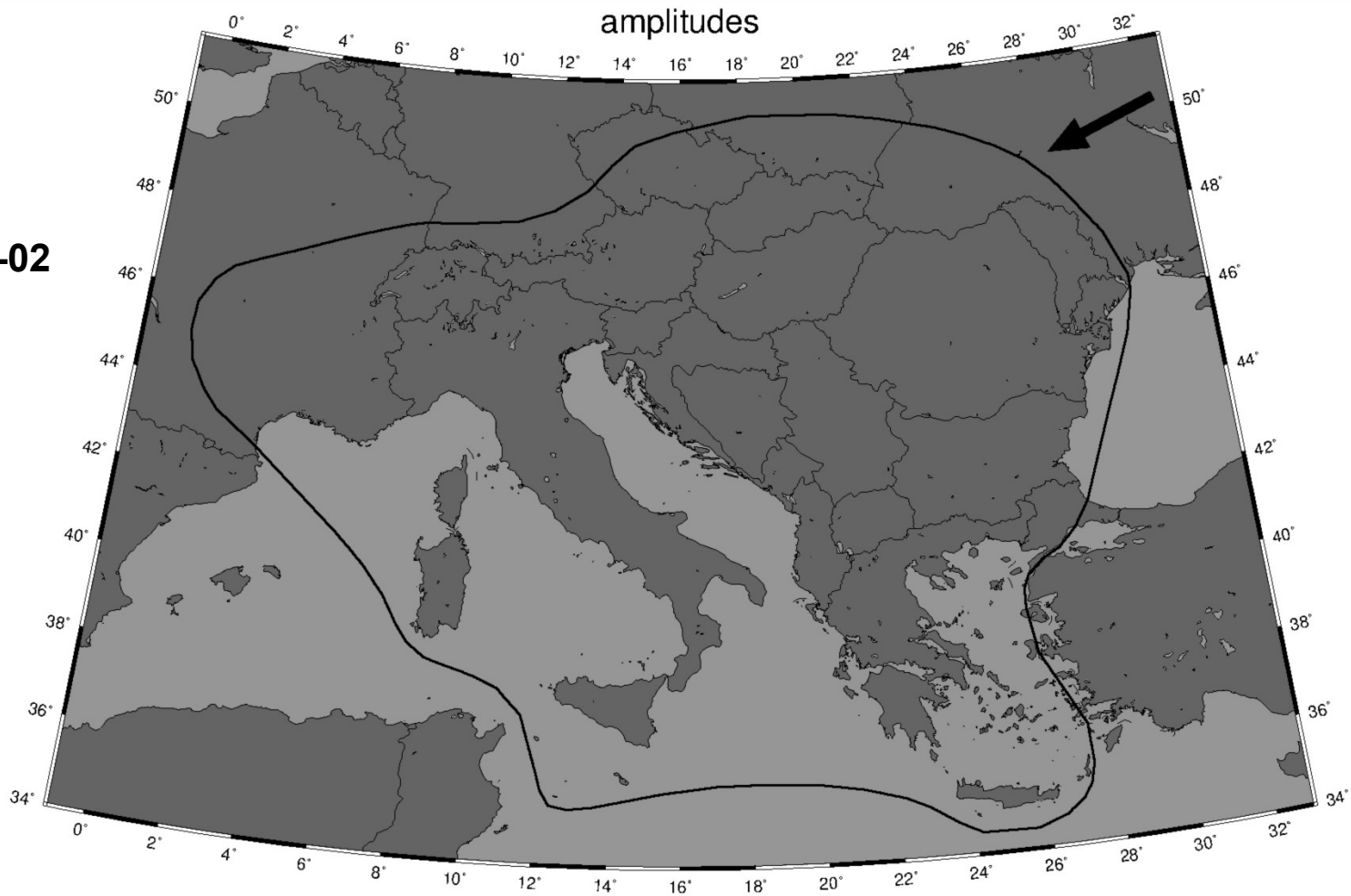
⁽²⁾ Institute for Geosciences, University of Kiel, Germany

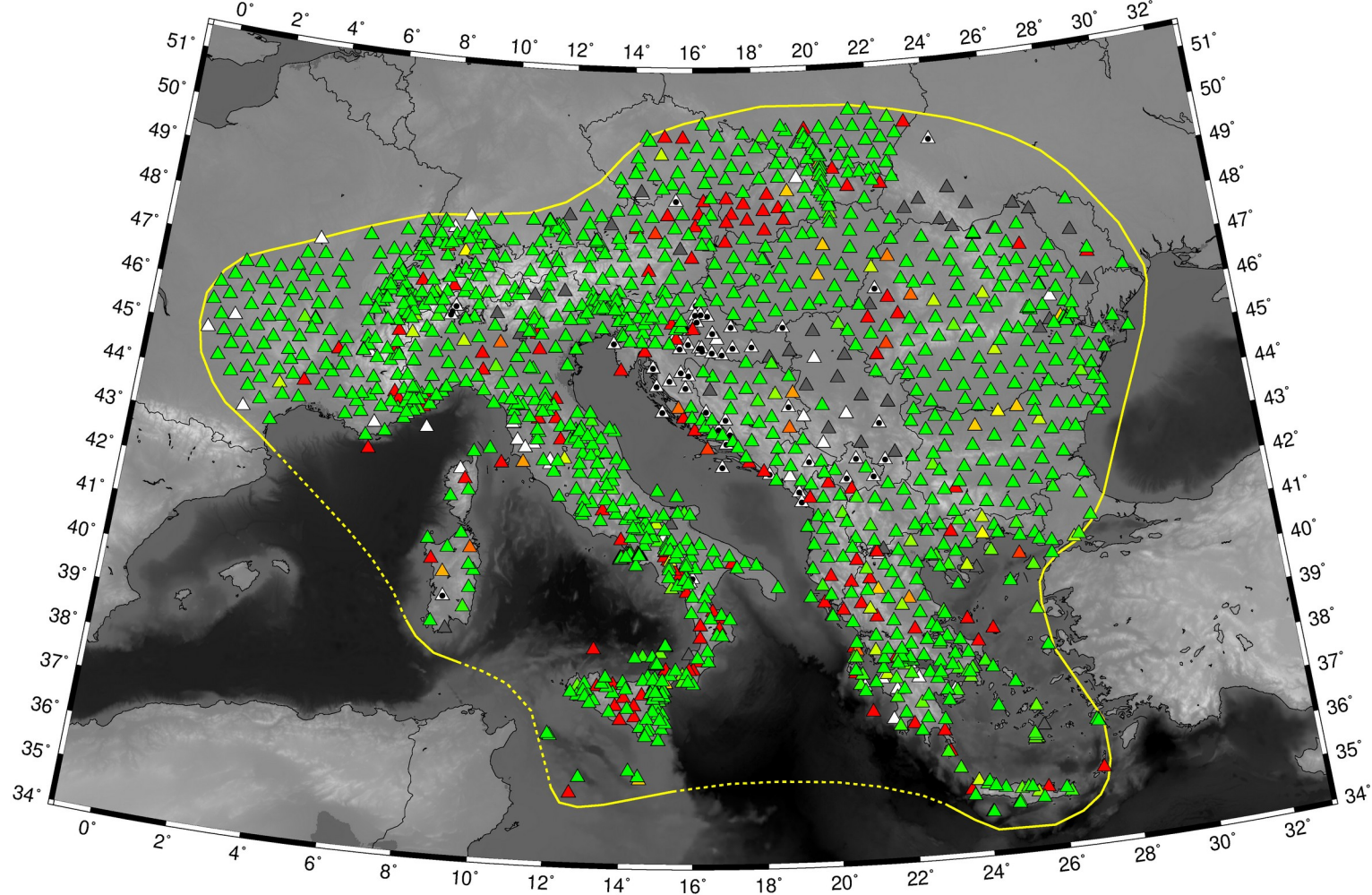
⁽³⁾ Department of Earth Sciences, Bullard Laboratories, University of Cambridge, Cambridge, United Kingdom

⁽⁴⁾ Department of Geophysics, Faculty of Science, University of Zagreb, Croatia

⁽⁵⁾ https://orfeus.readthedocs.io/en/latest/adria_array_main.html

Taiwan
M = 7.4
2024-04-02

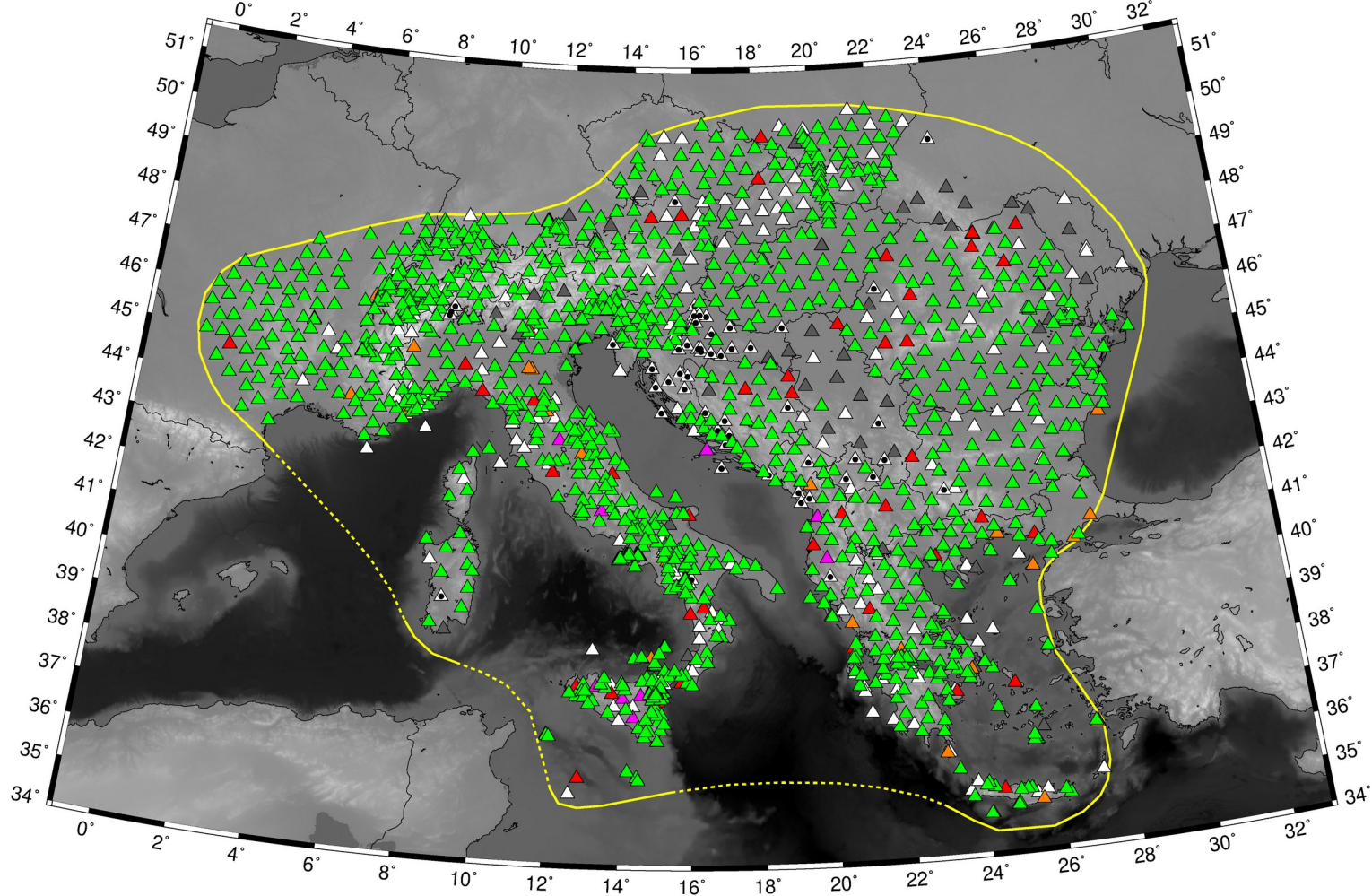




retrievability March 2024
tested in February 2025

0 10 20 30 40 50 60 70 80 90 100 percent

▲ metadata issue ▲ not available ▲ not in EIDA ▲ not operational



EQ data



no problem



warning



data issue



metadata issue



not available

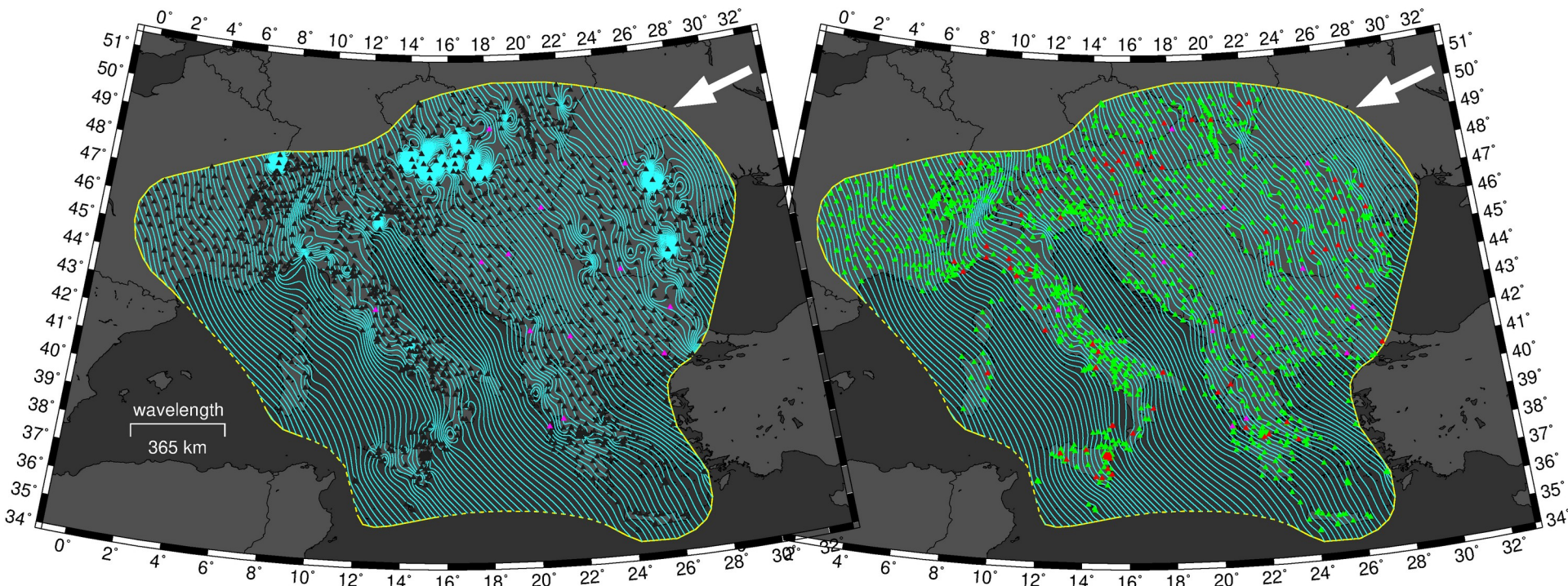


not in EIDA



not operational

WAVEFRONT TRACKING – Rayleigh waves, 90 s, fundamental mode



stations passing visual inspection



removed by visual inspection



no problem



removed by wavefront tracking



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